

Vivek Verma

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Education

- 2021 – at present  **Pursuing Ph.D., Department of Mathematics, Indian Institute of Technology Roorkee.**
CGPA- 8.5
Thesis title: *Developing new classical and quantum algorithms for multimedia data encryption.*
- 2017 – 2019  **M.Sc. in Mathematics and Computing, Department of Applied Mathematics, Indian Institute of Technology Dhanbad.**
CGPA- 8.93
- 2013 – 2016  **B.Sc. , Dr. Ram Manohar Lohia Awadh University, Ayodhya.**
Percentage- 62.22 %

PhD Thesis Title

-  Advancements in Classical and Quantum Algorithms for Secure Multimedia Data Encryption and Watermarking Supervisor: Dr. Sanjeev Kumar, Professor, Department of Mathematics, I.I.T. Roorkee
Expected Year of Award: 2026

Research Interests

-  Quantum Image Encryption, Image Encryption using Chaos Theory, Memristor, Image Edge Detection using Classical and Quantum Algorithms, Image watermarking, Compressive Sensing, Chaotic Dynamics and Applications, Memristor based Neural Networks Dynamics, Machine Learning, Post Quantum Cryptography.

Workshops and Conferences

- 2022  **RACAM**, Recent Advances in Computational analysis and Modelling, organized by Department of Mathematics, IIT Roorkee, June 20-24, 2022.
-  **Conference**, Dynamical systems, Control and Their Applications, organized by Department of Mathematics, IIT Roorkee, July 01-02, 2022.
- 2024  **Conference**, International Conference on Computations and Data Science (CoDS-2024), organized by Department of Mathematics jointly with iHUB DivyaSampark IIT Roorkee, March 08-10, 2024
-  **Conference** International Young Quantum Scientist's Meet 2024 organized by Department of Physics and Material Science and Engineering Jaypee Institute of Technology, Noida, India, September 16-20, 2024.
-  **Conference** International Conference on Applied Mathematical Sciences for Emerging Technologies (ICAMSET-2024) organized by Department of Mathematics, School of Engineering and Technology, Dhanalakshmi Srinivasan University, Perambalur, Tamil Nadu, India, December 13-14 2024.
-  **Conference** 5th International Conference on Mathematical Engineering and Managment Sciences (ICMEMS-2024) organized by Applied Science Cluster (Mathematics), School of Advanced Engineering (SoAE), UPES, Dehradun, India, December 13-14 2024.

Workshops and Conferences (continued)

- 2025
- **Conference** Third International Workshop on Advanced Topics in Mathematics (IWATM-2024) organized by the Centre for Applied Mathematics, International Institute of Information Technology Naya Raipur, during December 16 - 20, 2024.
 - **Workshop** Dynamical Systems and Chaos Theory: IoT and Cryptography Applications organized by the Department of Mathematics and Computing, Dr. B R Ambedkar National Institute of Technology, Jalandhar, Punjab, India under Global Initiative of Academic Networks (GIAN), January 06-11, 2025.
 - **Conference** International Symposium On Quantum Information and Communication (ISQIC), organized by TCG Centres for Research and Education in Science and Technology (TCG CREST), Kolkata, India, March 31- April 2, 2025.
 - **Conference** **Theory of Quantum Computation, Communication and Cryptography (TQC 2025)** organized by Indian Institute of Science and Colleges (IISC) Bangalore, India, under International Year of Quantum Science and Technology, September 13-19, 2025.
- 2026
- **Conference** **29th Annual Quantum Information Processing Conference (QIP 2026)** organized by University of Latvia, Riga and QIP 2026 organizing committee, Riga, Latvia, January 24-30, 2026
 - **Conference** International Young Quantum Scientist's Meet 2024 organized by Department of Physics and Material Science and Engineering Jaypee Institute of Technology, Noida, India, March 16-20, 2026.

Publications

- Verma, V., Kumar, S. and Rani, N. Novel image encryption algorithm using hybrid 3D-ICPCM and hessenberg decomposition. Nonlinear Dyn 112, 10583–10609 (2024). doi = 10.1007/s11071-024-09620-w.
- Verma, V., Kumar, S. Quantum image encryption algorithm based on 3D-BNM chaotic map. Nonlinear Dyn 113, 3829–3855 (2025) doi = 10.1007/s11071-024-10403-6
- Verma V., Tabekoueng Njitacke, Z., Kumar S. Quantum inspired image encryption using a memristive hopfield neural network. Journal of Information Security and Applications 100, 104464 (2026). doi =10.1016/j.jisa.2026.104464
- Verma V., Tabekoueng Njitacke, Z., Kumar S. Quantum inspired image encryption based on Neuron model and Power cube method. (Submitted)

Skills

- Languages ■ English, Hindi.
- Coding ■ Pytorch, Pennylane, Python, MATLAB, C++, Latex, Qiskit.
- Misc. ■ Leadership, Academic research, Teaching, Latex typesetting, and Publishing.

Awards and Achievements

- 2017
- Cleared IIT-JAM exam for M.Sc. program in IITs
 - Awarded MCM (Merit-Cum-Means)fellowship by IIT Dhanbad
- 2018
- Awarded MCM (Merit-Cum-Means)fellowship by IIT Dhanbad
- 2020
- Qualified CSIR(Council of Scientific and Industrial Research) NET (National Eligibility Test) for Lectureship, AIR-68
- 2023
- Global Quantum Scholarship for Womanium QAUNTUM 2023

Awards and Achievements (continued)

- 2024  **Trusted Reviewer in Institute of Physics (IOP) Publishing in Physica Scripta Journal, Journal Of Optics**
- 2026  **Travel Grant from Anusandhan National Research Foundation (ANRF), Govt. of India, to attend the 29th Annual Quantum Information Processing Conference (QIP 2026) held at the University of Latvia, Riga.**
 -  **Awarded Student Research Scholarship by the Quantum Information Processing (QIP 2026) organizing committee to attend the 29th Annual Quantum Information Processing Conference, University of Latvia, Riga.**

Teaching Experience

-  Graduate Teaching Fellow for **MAN-001, Mathematics I**, Department of Mathematics, IIT Roorkee (Spring 2022).
-  Graduate Teaching Fellow for **PMA-403, C++ Language**, Department of Mathematics, IIT Roorkee (Fall 2022).
-  Graduate Teaching Fellow for **MAN-002, Mathematics Methods**, Department of Mathematics, IIT Roorkee (Spring 2023).
-  Graduate Teaching Fellow for **MAN-001, Mathematics I**, Department of Mathematics, IIT Roorkee (Fall 2023).
-  Graduate Teaching Fellow for **MAI-102, Linear Algebra, Probability, and Statistics**, Department of Mathematics, IIT Roorkee (Spring 2024).
-  Teaching Assistant for the National Programme on Technology Enhanced Learning (NPTEL) Online Certification Course **Multivariable Calculus**, IIT Roorkee (Spring 2024).
-  Graduate Teaching Fellow for **PMA-403, C++ Language**, Department of Mathematics, IIT Roorkee (Fall 2024).
-  Teaching Assistant for the National Programme on Technology Enhanced Learning (NPTEL) Online Certification Course **Multivariable Calculus**, IIT Roorkee (Spring 2025).
-  Graduate Teaching Fellow for **PMA-403, C++ Language**, Department of Mathematics, IIT Roorkee (Fall 2025).
-  Teaching Assistant for the National Programme on Technology Enhanced Learning (NPTEL) Online Certification Course **Multivariable Calculus**, IIT Roorkee (Spring 2026).

References

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Professor Maheshanand

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